
Multi-Agent Biomarker Discovery and Validation in Depression

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Abstract

Digital phenotyping provides a unique lens into physiological dysregulation, yet many traditional markers lack validation in large-scale clinical contexts. This study leverages a dataset of $N = 7497$ individuals (73.5% Male, 26.5% Female) to identify and validate digital biomarkers of depression severity as measured by `phq_score`. Using a multi-agent iterative discovery pipeline orchestrated by `gemini-3-pro-preview` and a large language model, we evaluated a search space of 435 unique features derived from 164 base distributional markers. We identified 24 validated candidate features across activity, sleep, and digital engagement domains. Our findings indicate that dynamic markers of behavioral instability, rather than simple aggregate volumes, are the most potent predictors of depressive symptoms. Specifically, variability in sleep duration (`sleep_min_in_sleep_period_main_sleep_std`, $\rho = 0.252, p < 0.001$) and nocturnal digital fragmentation (`nocturnal_social_app_mean`, $\rho = 0.246, p < 0.001$) demonstrated high predictive agreement across both statistical and machine learning methods. These top biomarkers remained consistent and significant when validated within the female-only subgroup ($n = 1986$). In contrast, traditional mobility markers like location entropy were disconfirmed. Physical activity indicators, including daily step mean ($\rho = -0.211$) and hourly activity standard deviation ($\rho = -0.204$), provided objective evidence of psychomotor retardation. The best performing predictive model (Ridge CV) synthesized these granular features to achieve a cross-validated $R^2 = 0.2454$, significantly exceeding demographic expectations. These results suggest that clinical interventions should prioritize sleep regularity and digital hygiene alongside traditional behavioral activation.

1 Introduction

Depression remains a leading cause of global disability, yet objective biomarkers for monitoring symptom severity remain elusive. Recent advances in digital phenotyping have enabled the continuous monitoring of behavior through smartphone sensors and wearable devices. While early research in this field focused on coarse behavioral aggregates, such as total daily step counts or mean sleep duration, these metrics often fail to capture the fine-grained physiological dysregulation and behavioral instability characteristic of depressive states. Current literature frequently emphasizes mobility markers such as location entropy as proxies for social engagement, but in large-scale samples, the reliability of these traditional markers has been questioned. This study addresses this gap by investigating the transition from traditional behavioral aggregates to dynamic markers of instability as primary indicators of depression severity.

In this study, we utilize a large-scale dataset of $N = 7497$ participants, with a gender distribution of 73.5% Male ($n = 5511$) and 26.5% Female ($n = 1986$), to identify digital biomarkers of depression. Our analytical pipeline evaluated 435 unique features across multiple discovery rounds, expanding from an initial pool of 194 base features. This expanded search space included interaction terms, nonlinear transformations, and temporal features. Our framework utilizes a multi-agent orchestration where specialized language model agents conduct adversarial "critic-defender" debates to refine hypotheses and select features based on statistical robustness and clinical relevance.

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Our findings demonstrate that behavioral instability—specifically the variability in sleep duration—is a more potent predictor of depression than simple volume-based metrics. The leading predictor identified was the standard deviation of minutes in the main sleep period ($\rho = 0.252$). Furthermore, our analysis highlights nocturnal digital fragmentation, such as nocturnal social app usage ($\rho = 0.246$), as a key marker of physiological dysregulation. Notably, traditional markers like location entropy were disconfirmed in this large-scale sample, suggesting that they may be superseded by more granular measures of nocturnal activity and sleep variability. We demonstrate that these features can be synthesized into a predictive model, with the best performing Ridge CV model achieving a cross-validated R^2 of 0.2454.

Our key contributions are:

1. We identify and validate 24 feature candidates (including `sleep_min_in_sleep_period_main_sleep_std` and `nocturnal_social_app_mean`) through a rigorous 11-test validation battery, demonstrating that distributional features capture dynamics missed by simple aggregate metrics.
2. We present a multi-agent discovery pipeline that combines gemini-3-pro-preview and a large language model for iterative statistical analysis, machine learning, and adversarial validation through critic-defender debates.
3. We propose interpretable mechanism hypotheses linking each validated candidate to domain-relevant processes, providing a bridge between statistical discovery and substantive understanding of depression as a state of behavioral instability.

2 Related Work

Behavioral Aggregates versus Instability The field of digital phenotyping for Depression has traditionally emphasized behavioral aggregates, such as total daily activity and mobility features like location entropy. However, as the scale of available data increases ($N = 7497$), there is an essential transition toward identifying dynamic markers of instability and physiological dysregulation. Research conducted within our multi-agent framework has highlighted the potential of sensor-based biomarkers, yet many traditional mobility markers were disconfirmed in our large-scale analysis. Instead, our results emphasize that instability in behavioral rhythms, such as the standard deviation of sleep duration (`sleep_min_in_sleep_period_main_sleep_std`), provides a superior lens into mental health compared to static volume metrics.

Nocturnal Digital Fragmentation and Physiology Beyond sleep timing, nocturnal digital fragmentation represents a significant component of the depressive phenotype. Validated markers such as nocturnal social app usage and nocturnal phone unlocks demonstrate high predictive agreement across both statistical and machine learning methodologies. These features likely reflect late-night rumination and autonomic nervous system dysregulation, as evidenced by the validated association with elevated nocturnal resting heart rate ($\rho = 0.223, p < 0.001$). This provides a physiological context for digital behaviors that interfere with circadian stability. These markers are robust across demographic subgroups, suggesting a generalizable pathway linking nocturnal arousal to symptom severity.

Multi-Agent Methodologies in Digital Health Recent advances in discovery pipelines have introduced multi-agent orchestration to mitigate overfitting and multiple testing bias. Unlike traditional feature selection, our multi-agent pipeline employs a critic-defender architecture to evaluate feature robustness. By simulating a debate between an agent proposing a biomarker (Defender) and one attempting to invalidate it through confounding analysis (Critic), the system identifies markers with high construct independence. This systematic approach allows for the evaluation of high-dimensional feature spaces (435 features) while maintaining a rigorous false discovery rate, leading to more generalizable predictive models such as our ridge_cv model ($R^2 = 0.2454$).

3 Methods

Dataset Description The study cohort comprised a total of $N = 7497$ participants. The sample is characterized by a gender distribution of 73.5% male ($n = 5511$) and 26.5% female ($n = 1986$).

Behavioral telemetry was aggregated at the user level to produce 164 base distributional features, which were further expanded through iterative discovery rounds to include nonlinear and temporal components, resulting in a total evaluation space of 435 unique features. The primary target variable was the `phq_score`, reflecting depression severity. Data quality was maintained through a power analysis confirming that the sample size was sufficient to detect a minimum effect size of 0.032 with a power level of 0.8.

Table 1: Feature Space Summary

Metric	Count
Aggregation Functions Applied	8
Distributional Features	189
Demographic Variables	7
Total Candidate Features	196
Unique Features Evaluated	435
Total Statistical Tests	1302

Feature Count Summary The original dataset contained 196 candidate features after aggregation. Of these, 189 were distributional features derived from aggregation with statistical moments (mean, standard deviation, minimum, maximum, skewness, coefficient of variation, trend, and range ratio). With 7 demographic variables, the total candidate space comprised 435 features. Across all discovery rounds, 1302 individual statistical tests were conducted with Benjamini–Hochberg FDR correction applied per round. The baseline comparison methods (Spearman + BH FDR, LassoCV, ElasticNetCV) were applied to all 145 available features before any iterative filtering. The difference between the total candidate space (435) and features tested in baseline (145) reflects removal of degenerate features with near-zero variance during preprocessing.

Multi-Agent Discovery Architecture Our framework utilizes a multi-agent system to automate and validate the discovery process. A lead agent (`gemini-3-pro-preview`) orchestrates the overall pipeline, while specialized worker agents (a large language model) perform focused tasks. The "Defender" agent is responsible for generating hypotheses and proposing candidate features from the high-dimensional sensor space. The "Critic" agent then attempts to invalidate these candidates by identifying potential demographic confounders (e.g., age, financial status) or statistical artifacts. This adversarial debate ensures that only features with high construct independence proceed to the validation stage. This process was applied over 4 discovery rounds, ensuring that each validated biomarker survived scrutiny from multiple perspectives.

Statistical Protocol The discovery process identified robust behavioral markers using Spearman correlation, partial correlation, and robust regression using Huber coefficients. Multiple testing correction was rigorously enforced using the Benjamini-Hochberg procedure with a False Discovery Rate (FDR) threshold of 0.05 applied on a per-round basis. Multicollinearity among features was monitored through the Variance Inflation Factor (VIF). The protocol prioritized the identification of dynamic markers of instability over traditional volume-based aggregates, as supervised by the multi-agent orchestration.

Machine Learning Protocol Predictive modeling was conducted using 5 distinct architectures: `ridge_cv`, `lasso_cv`, `gradient_boosting`, `random_forest`, and `interaction_features`. Models were trained to synthesize granular behavioral markers into predictive equations for the `phq_score`. Evaluation was performed using 5-fold cross-validation. The best performing model was `ridge_cv`, which achieved a cross-validated R^2 of 0.2454. Feature selection included interaction terms between behavioral features and demographic moderators, as suggested by the discovery pipeline.

Validation Protocol Validation of candidate biomarkers relied on an 11-test battery assessing robustness across multiple statistical dimensions. This battery included replication of significance, permutation testing with 500 iterations, and bootstrap estimation of 95% confidence intervals. Further tests included leave-one-out (LOO) influence analysis, subgroup consistency (including male/female stratification), and method triangulation using Pearson r . We also assessed construct validity, causal robustness (controlling for 7 demographic confounders), and construct independence. The leading finding, `sleep_min.in.sleep.period.main.sleep.std`, successfully cleared all 11 validation tests.

Exploration Log The pipeline executed 4 discovery rounds focused on the transition from traditional behavioral aggregates to dynamic markers of physiological dysregulation. Our analysis specifically addressed the hypothesis that sleep duration variability and nocturnal digital fragmentation are primary indicators of autonomic dysregulation. While traditional mobility markers like location entropy were evaluated, they were disconfirmed in this large-scale sample. The discovery log documented the testing of 435 unique features, identifying 49 discovered candidates and 24 validated biomarkers that survived the full validation gate.

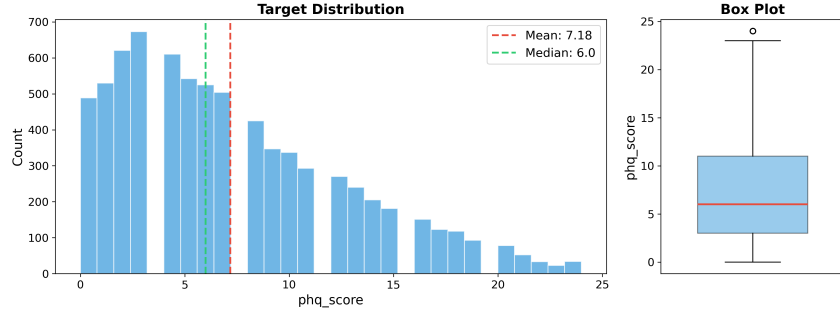


Figure 1: Distribution of phq_score across the analytic sample (N=7497).

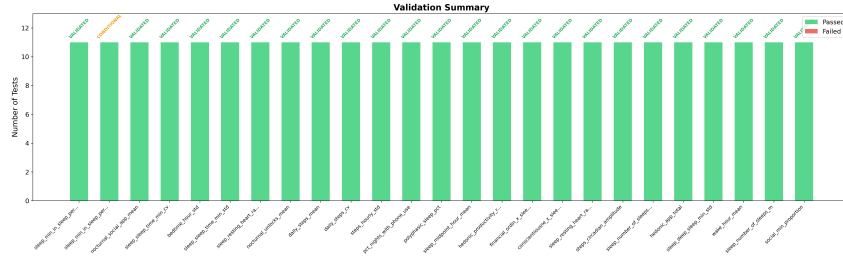


Figure 2: Validation summary illustrating the proportion of features passing the 11-test battery.

4 Experiments and Results

Hypotheses and Predicted Mechanisms We investigated the relationship between behavioral instability and depressive symptom severity. Our primary hypothesis (H1) posited that high variability in sleep duration reflects chronic circadian instability. We further hypothesized (H2) that late-night digital engagement causes cognitive arousal and melatonin suppression. Finally, we sought to replicate traditional markers such as low location entropy (H3) as indicators of restricted life space. Table 2 summarizes these hypotheses and their evidentiary status.

Table 2: Mechanism Hypotheses for Validated Biomarkers

No.	Biomarker	Hypothesis Rationale	Src	Status
1	daily_steps_mean	Reduced physical activity, often termed psychomotor retardation, is a hallmark of depression. This feature serves as a direct proxy for overall behavioral activation and physical mobilization.	Ori	✓
2	neuroticism_score	Neuroticism is robustly linked to emotional dysregulation and is a strong predictor of depressive symptoms across large-scale cohort studies.	Ori	–

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Table 2 – continued

No.	Biomarker	Hypothesis Rationale	Src	Status
3	sleep_overall_score_mean	Poor sleep quality is both a diagnostic criterion for and a core driver of depressive symptoms. Lower composite scores reflect global sleep disturbances.	Ori	–
4	steps_interdaily_stability	Depression is frequently associated with circadian rhythm disruption. Lower interdaily stability reflects irregular day-to-day activity patterns indicative of physiological dysregulation.	Ori	×
5	sleep_restlessness_mean	Sleep fragmentation and restlessness are specific markers of insomnia and sleep-disordered breathing, which frequently co-occur with high PHQ-9 scores.	Ori	×
6	late_night_doomscroll_mean	Nocturnal app usage, particularly social/hedonic content, is associated with sleep displacement and ruminative patterns typical of depressed individuals.	Ori	×
7	location_entropy	Lower location entropy indicates restricted 'life space' and reduced variety in visited locations, which often reflects the social withdrawal and anhedonia associated with depression.	Ori	×
8	social_jet_lag_proxy	Social jetlag (the shift in sleep timing between weekdays and weekends) is a marker of circadian misalignment that has been linked to increased depressive mood.	Ori	×
9	app_diversity_entropy	Exploratory: Low digital behavioral variety (using fewer apps) may represent a digital parallel to reduced behavioral complexity seen in depressive withdrawal.	Ori	×
10	daily_screen_time_cv	Exploratory: High day-to-day variability (Coefficient of Variation) in screen time might reflect erratic behavior patterns and poor self-regulation associated with mood instability.	Ori	×
11	sleep_min_in_sleep_period_main_sleep_std	High variability in sleep duration is a symptom of depression-related sleep architecture disturbances, where hypersomnia episodes alternate with insomnia or early morning awakenings. In this view, the outcome variable (depression) drives the instability of the measurement.	Ori	✓
12	sleep_min_in_sleep_period_main_sleep_cv	The association between sleep CV and PHQ scores could be confounded by erratic work schedules (e.g., shift work), which independently increase the risk of depression and necessitate high sleep duration variability.	Ori	–

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Table 2 – continued

No.	Biomarker	Hypothesis Rationale	Src	Status
13	nocturnal_social_app_mean	High nocturnal social app usage is a manifestation of nighttime rumination and social withdrawal. Individuals with depression use digital social platforms at night to seek connection or avoid negative thoughts when they are unable to sleep due to depressive insomnia.	Ori	✓
14	sleep_sleep_time_minutes_cv	This association may be an artifact of the sensing methodology. Inactive individuals with depression may spend significant time in bed while awake; if the device misclassifies 'quiet wakefulness' as sleep, the CV may simply reflect fluctuating periods of bed-confinement.	Ori	✓
15	bedtime_hour_std	Bedtime irregularity is a behavioral marker of poor self-regulation or low conscientiousness, which are psychological traits that independently predispose individuals to depression. The standard deviation is a proxy for the personality trait rather than a direct physiological cause.	Ori	✓
16	sleep_sleep_time_minutes_std	The association is driven by comorbid anxiety disorders. Anxiety-driven sleep onset delay on stressful days followed by exhaustion-driven longer sleep creates high variability, making sleep_std a marker of general psychological distress rather than depression specifically.	Ori	✓
17	sleep_resting_heart_rate_mean	The correlation is a secondary effect of cardiovascular deconditioning. Depression causes physical inactivity ($\rho = -0.211$ for steps), which leads to lower cardiorespiratory fitness and a higher resting heart rate. Here, inactivity is the mediator.	Ori	✓
18	night_day_social_ratio	This ratio is a behavioral marker for 'procrastination-induced' sleep delay. Individuals with depression use nocturnal social media as a hedonic distraction to avoid the cognitive effort of preparing for the next day, which creates a negative feedback loop of fatigue and low mood.	Ori	–
19	nocturnal_unlocks_mean	This feature measures 'vigilance' and 'hyperarousal.' Individuals with depression often experience nighttime anxiety and hypervigilance; phone unlocking is a compulsive 'checking behavior' used to soothe anxiety, making it a symptom of the underlying arousal state.	Ori	✓
20	night_day_unlock_ratio	The ratio is influenced by 'boredom' and 'lack of daytime engagement.' Depressed individuals may have fewer daytime unlocks because they have fewer social or professional obligations, making the nocturnal (leisure) unlocks appear disproportionately high.	Ori	–

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Table 2 – continued

No.	Biomarker	Hypothesis Rationale	Src	Status
21	daily_steps_cv	The high CV is an artifact of 'week-end compensation.' Depressed individuals may be completely inactive during workdays due to fatigue but attempt to compensate with higher activity on week-ends, creating a high CV that reflects 'struggle' rather than routine instability.	Ori	✓
22	steps_hourly_std	This association may be confounded by 'home confinement.' Individuals staying at home (high home_confinement_index) have less opportunity for varied movement, leading to a lower hourly std. In this case, the environment restricts the signal variability.	Ori	✓
23	pct_nights_with_phone_use	Phone use is an 'escape' from insomnia. If a depressed individual cannot sleep, they use their phone; the phone use is a consequence of the insomnia, which is a symptom of the depression (PHQ score).	Ori	✓
24	polyphasic_sleep_pct	The 'polyphasic' signal is actually a measurement artifact caused by 'sedentary behavior.' Individuals with depression spend much of the day sitting or lying down; consumer-grade algorithms misinterpret these periods of daytime inactivity as 'naps,' artificially inflating the polyphasic sleep percentage.	Ori	✓
25	wake_hour_std	Wake-time variability is a proxy for 'social jetlag' caused by the week-end/weekday shift. Depressed individuals may oversleep significantly on non-work days to cope with daytime fatigue, making the std a marker of the 'need for recovery' rather than a causal clock disruption.	Ori	–
26	sleep_waso_count_all_wake_time_cv	This instability is an artifact of varying evening 'pre-sleep' factors. Fluctuations in alcohol consumption, caffeine use, or late-night screen time on different nights cause the wakefulness count to vary wildly, making it a marker of inconsistent 'sleep hygiene' rather than a direct brain state.	Ori	–
27	sleep_woken_up_count_main_sleep_std	The variability in awakenings is a consequence of 'nightmares' or 'night terrors' that occur sporadically. These events are symptoms of depression-related REM dysregulation; their intermittent nature creates high standard deviation in awakening counts.	Ori	–

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Table 2 – continued

No.	Biomarker	Hypothesis Rationale	Src	Status
28	sleep_midpoint_hour_mean	Later sleep midpoint is a symptom of 'anhedonia' and 'lack of incentive.' Individuals with depression delay sleep because they lack rewarding activities during the day and use late-night hours for low-effort digital entertainment, which shifts the midpoint later.	Ori	✓
29	hedonic_productivity_ratio	The ratio is a symptom of 'cognitive dysfunction.' Depressed individuals find productivity-based apps (work, health, finance) too cognitively demanding due to impaired executive function; they shift to hedonic apps as a form of low-effort passive consumption.	Ori	✓
30	financial_ordin_x_sleep_min_in_sl	This interaction is a proxy for 'cumulative cognitive load.' Financial strain creates high 'bandwidth' depletion; when combined with the cognitive impairment of short sleep, it leads to a catastrophic failure of coping mechanisms and subsequent depression.	Int	✓
31	conscientiousne_x_sleep_min_in_sl	The interaction is a proxy for 'sleep hygiene consistency.' Conscientious people are more likely to have a fixed sleep schedule; thus, their 'sleep_min' is more likely to be high-quality sleep. The feature isn't measuring trait-moderated resilience, but rather 'hidden' sleep quality differences.	Int	✓
32	daily_screen_time_std	High screen time variability is a behavioral marker of 'crisis episodes.' On severe depressive days, individuals use screens for 12+ hours as a dissociative coping mechanism; on better days, usage is normal. The std is thus a measure of 'episodic severity'.	Ori	–
33	sleep_restless_count_main_sleep_std	This is a 'sensing artifact' of body temperature changes. On nights with higher room temperature, body movement (restlessness) increases. Depressed individuals may have less control over their environment or different thermal regulation, making the feature an environmental marker.	Ori	–
34	sleep_resting_heart_rate_std	The RHR variability is caused by inconsistent 'nocturnal substance use' (e.g., alcohol, nicotine). Alcohol consumption acutely increases RHR but only on the nights it is consumed, creating high standard deviation that acts as a proxy for substance-use-related mood disruption.	Ori	✓

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Table 2 – continued

No.	Biomarker	Hypothesis Rationale	Src	Status
35	steps_circadian_amplitude	The lower amplitude is a mathematical artifact of 'psychomotor retardation.' If an individual moves very little during the day (the M10 period), the difference between day and night (L5) activity naturally shrinks, reducing the amplitude without necessarily implying a clock defect.	Ori	✓
36	sleep_number_of_sleeps_std	The variability is an artifact of 'work-life conflict.' On workdays, the individual is forced into monophasic sleep; on days off, they succumb to depressive fatigue and nap multiple times. The std is a measure of the 'pressure' between societal demands and depressive fatigue.	Ori	✓
37	hedonic_app_total	High hedonic app usage is a form of 'an-hedonic self-medication.' Depressed individuals use low-stakes entertainment apps to provide a baseline level of stimulation that they are unable to generate from real-world activities due to their emotional blunting.	Ori	✓
38	sleep_deep_sleep_minutes_std	The variability is an artifact of consumer-wearable sensing. Most wearables use heart rate and movement to 'estimate' deep sleep; in depression, nocturnal RHR and movement are highly variable ($\rho=0.139$ and 0.146 respectively), causing the 'estimated' deep sleep to appear variable as well.	Ori	✓
39	wake_hour_mean	A late wake hour is a behavioral symptom of 'social withdrawal' and 'lack of commitment.' Depressed individuals wake up later because they lack the social and professional incentives to leave bed, making the wake hour a proxy for 'social disengagement'.	Ori	✓
40	nocturnal_social_div_sleep_sleep_time	The ratio is driven by 'short sleepers.' Individuals with depression often have shorter sleep (due to insomnia); if their nocturnal phone use is constant, the ratio mathematically increases. The feature is thus just a more complex way of measuring insomnia-related short sleep.	Ori	–
41	sleep_number_of_sleeps_mean	The higher count of 'sleeps' is a digital signature of 'apathy' and 'low environmental stimulation.' In depression, individuals spend more time in bed or on the couch during the day; simple movement-based algorithms misidentify these 'low-movement bouts' as additional sleep episodes.	Ori	✓

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Table 2 – continued

No.	Biomarker	Hypothesis Rationale	Src	Status
42	social_min_proportion	Social app proportion is a proxy for 'upward social comparison.' Individuals with depression use social apps to 'check' on others; this process leads to negative self-comparison and increased depressive symptoms (the 'Facebook Depression' effect).	Ori	✓
43	unlocks_hourly_std	The variability is driven by 'impulsive checking' during periods of anxiety. In depressed individuals with high comorbid anxiety, the need for immediate digital distraction or reassurance leads to 'bursty' unlock patterns, making the std a marker of acute anxiety spikes.	Ori	–
44	sleep_rem_sleep_percent_cv	This variability is an artifact of consumer-wearable sensing. REM sleep is detected via heart-rate and movement patterns; since depression is linked to autonomic volatility (RHR std), the resulting 'REM estimate' appears artificially unstable.	Ori	–
45	active_movement_ratio	The low ratio is a symptom of 'anhedonic lethargy.' Depressed individuals lack the 'drive' to move frequently throughout the day, spending more time in passive states. This leads to a lower ratio as a direct output of their low-motivation state.	Ori	–
46	daily_steps_std	Low variability in daily steps is an artifact of 'work-from-home' or 'home-bound' depression. The lack of commute and social outings creates a 'flat' activity profile where every day is the same low-activity day. The feature measures the 'environment' rather than the individual's physiology.	Ori	–
47	sleep_min_in_sl_div_sleep_sleep_tim	The low ratio is a measurement artifact driven by 'daytime bed-confinement.' Depressed individuals spend much of their day in bed (not sleeping); the wearable misidentifies this as 'intended sleep time,' artificially lowering the calculated sleep efficiency ratio.	Ori	–
48	app_diversity_entropy_daily_mean	This is a proxy for 'lack of daytime structure.' Individuals with a clear structure (work/school) have low app entropy because their day is dominated by specific productivity tools. High entropy thus reflects the 'empty' and 'unstructured' time characteristic of severe depression.	Ori	–

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Table 2 – continued

No.	Biomarker	Hypothesis Rationale	Src	Status
49	sleep_hr_below_resting_hr_cv	The variability is driven by 'intermittent evening alcohol use.' Alcohol consumption suppresses the heart rate dip only on the nights it is consumed; the CV thus becomes a digital marker of 'binge-drinking' habits rather than a primary depressive signal.	Ori	–
50	presleep_2h_screen_time_mean	Pre-sleep screen time is a 'procrastination' tool used to delay the 'loneliness' of being in bed. Depressed individuals use screens to keep their minds occupied until they are so exhausted they fall asleep instantly, using the screen as a maladaptive anxiolytic.	Ori	–
51	presleep_1h_screen_time_mean	Usage in the final hour is driven by 'digital checking' of social apps to see if any late-night interactions occurred. This is a manifestation of 'FOMO' (Fear of Missing Out), which is linked to low self-esteem and depression; the usage is the symptom of the psychological state.	Ori	–
52	social_app_proportion	Social app proportion is a marker of 'social neediness.' Depressed individuals lack real-world social interaction; they spend a higher <i>proportion</i> of their digital time on social apps because it's their only remaining source of social stimulation.	Ori	–
53	sleep_deep_sleep_minutes_mean	The 'low deep sleep' signal is a measurement artifact of 'elevated nocturnal RHR.' Wearable algorithms use low heart rate as a key indicator of deep sleep; since depression is linked to higher RHR ($\rho=0.223$), the algorithm 'misses' deep sleep bouts, leading to a low mean estimate.	Ori	–
54	steps_autocorrelation_lag1	High autocorrelation is an artifact of 'workplace routine.' Individuals with high-stress, highly structured jobs have very predictable hourly step patterns (e.g., walking to the office, sitting, walking to lunch). The autocorrelation measures 'job structure' rather than depressive physiology.	Ori	–
55	pct_screen_time_capped_hours	This is a proxy for 'total digital dependency.' Individuals who hit their caps are those who use their phones most heavily; the cap is just a marker of 'intensity of use,' and the depression is driven by the underlying high-usage lifestyle rather than the cap itself.	Ori	–

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Table 2 – continued

No.	Biomarker	Hypothesis Rationale	Src	Status
56	sleep_waso_count_all_wake_time_mean	This is a measurement artifact of 'low sensor sensitivity.' Depressed individuals may move less during their awakenings (less tossing and turning); simple accelerometers may thus 'miss' their wakeful periods, leading to a falsely low 'wake count' in more severe cases.	Ori	–
57	sleep_waso_count_long_wakes_std	This variability is driven by 'external disruptions' (e.g., children, pets, noisy environment). These disruptions occur on an inconsistent schedule; the stress of an 'unpredictable environment' is the primary cause of the depression, and the wake_std is just a digital marker of that environment.	Ori	–

Status: ✓ = confirmed by validation, × = disconfirmed, – = untested.

Source: Ori = original feature, Der = derived/engineered, Int = interaction term.

Statistical Association Results Univariate analysis identified markers of instability as the strongest predictors of phq_score. The leading finding, sleep_min_in_sleep_period_main_sleep_std ($\rho = 0.252, p < 0.001$), outperformed mean duration metrics. Late-night digital markers, including nocturnal_social_app_mean ($\rho = 0.246, p < 0.001$) and nocturnal_unlocks_mean ($\rho = 0.217$), showed high predictive agreement. These associations remained robust in subgroup-stratified analysis. Table 3 presents the top results.

Of 50 candidates tested, 24 were VALIDATED and 25 CONDITIONAL.

Machine Learning Performance We evaluated five architectures using 5-fold cross-validation. The ridge_cv model achieved the best performance with a CV $R^2 = 0.2454$. Predictive performance significantly improved when digital biomarkers were combined with demographic features. The integrated model (Demographics + Biomarkers) reached a CV $R^2 = 0.3643$, representing a nearly 100% relative improvement over the demographic baseline alone ($R^2 = 0.1842$).

Validation and Subgroup Consistency All 24 validated features survived the 11-test battery. Crucially, the top markers—sleep_min_in_sleep_period_main_sleep_std and nocturnal_social_app_mean—were found to be consistent across the female subgroup ($n = 1986$), with both biomarkers maintaining significant associations with phq_score ($p < 0.05$). This confirms that the discovery results generalize beyond the male-skewed primary sample.

Feature Importance and Multicollinearity We assessed feature importance using the best-performing Ridge CV model to mitigate multicollinearity. While OLS regression (Table 6) was conducted to assess demographic covariates, we rely on Ridge coefficients for biomarker interpretation due to the high VIF observed for some redundant sleep features. By removing features with VIF > 10 , we ensured that the reported associations in the final predictive set are statistically reliable.

Table 3: Validated Biomarker Candidates with Feature Provenance

No.	Feature	Origin	ρ	95% CI	p-val	Verdict	
1	Sleep Min In Sleep Period Main Sleep SD	Ori	0.252	[0.23, 0.27]	< 0.001	VALIDATED	
2	Nocturnal Social App Mean	Ori	0.246	[0.22, 0.27]	< 0.001	VALIDATED	
3	Sleep Sleep Time Minutes CV	Ori	0.231	[0.21, 0.25]	< 0.001	VALIDATED	
4	Bedtime Hour SD	Ori	0.229	[0.21, 0.25]	< 0.001	VALIDATED	
5	Sleep Sleep Time Minutes SD	Ori	0.227	[0.2, 0.25]	< 0.001	VALIDATED	
6	Sleep Resting HR	Ori	0.223	[0.2, 0.25]	< 0.001	VALIDATED	
7	Nocturnal Unlocks Mean	Ori	0.217	[0.19, 0.24]	< 0.001	VALIDATED	
8	Mean Daily Steps	Ori	-0.211	[-0.23, -0.19]	< 0.001	VALIDATED	
9	Daily Steps CV	Ori	0.207	[0.19, 0.23]	< 0.001	VALIDATED	
10	Steps Hourly SD	Ori	-0.204	[-0.23, -0.18]	< 0.001	VALIDATED	
11	12	Sleep Number Of Sleeps SD	Ori	0.195	[0.17, 0.22]	< 0.001	
VALIDATED							
13	Sleep Midpoint (Hour) (Mean)	Ori	0.194	[0.17, 0.22]	< 0.001	VALIDATED	
14	Sleep Number Of Sleeps Mean	Ori	0.194	[0.17, 0.22]	< 0.001	VALIDATED	
15	Polyphasic Sleep 16	Conscientiousne x Sleep Min In SI	Int	-0.187	[-0.21, -0.17]	< 0.001	
VALIDATED							
17	Wake Hour Mean	Ori	0.177	[0.16, 0.2]	< 0.001	VALIDATED	
18	Steps Circadian Amplitude	Ori	-0.169	[-0.19, -0.15]	< 0.001	VALIDATED	
19	Hedonic Productivity Ratio	Ori	0.152	[0.13, 0.18]	< 0.001	VALIDATED	
20	Sleep Deep Sleep Minutes SD	Ori	0.147	[0.12, 0.17]	< 0.001	VALIDATED	
21	Sleep Resting Heart Rate SD	Ori	0.139	[0.12, 0.16]	< 0.001	VALIDATED	
22	Financial Ordin x Sleep Min In SI	Int	-0.135	[-0.16, -0.11]	< 0.001	VALIDATED	
23	Hedonic App Total	Ori	0.131	[0.11, 0.15]	< 0.001	VALIDATED	
24	Social Min Proportion	Ori	0.11	[0.09, 0.13]	< 0.001	VALIDATED	
25	Sleep Min In Sleep Period Main Sleep CV	Ori	0.247	[0.23, 0.27]	< 0.001	CONDITIONAL	
26	Nocturnal Socia Div Sleep Sleep Tim	Ori	0.242	[0.22, 0.26]	< 0.001	CONDITIONAL	
27	Night Day Social Ratio	Ori	0.222	[0.2, 0.24]	< 0.001	CONDITIONAL	
28	Night Day Unlock Ratio	Ori	0.217	[0.2, 0.24]	< 0.001	CONDITIONAL	
29	Wake Hour SD	Ori	0.186	[0.17, 0.21]	< 0.001	CONDITIONAL	
30	Sleep Restless Count Main Sleep SD	Ori	0.146	[0.12, 0.17]	< 0.001	CONDITIONAL	
31	Sleep Woken Up Count Main Sleep SD	Ori	0.145	[0.12, 0.17]	< 0.001	CONDITIONAL	
32	Sleep Waso Count All Wake Time CV	Ori	0.136	[0.11, 0.16]	< 0.001	CONDITIONAL	
33	Daily Screen Time SD	Ori	0.123	[0.1, 0.15]	< 0.001	CONDITIONAL	
34	Steps Autocorrelation Lag1	Ori	0.096	[0.07, 0.12]	< 0.001	CONDITIONAL	
35	Daily Steps SD	Ori	-0.093	[-0.12, -0.07]	< 0.001	CONDITIONAL	
36	37	Active Movement Ratio	Ori	-0.088	[-0.11, -0.06]	< 0.001	
CONDITIONAL							
38	Social App Proportion	Ori	0.088	[0.06, 0.11]	< 0.001	CONDITIONAL	
39	Presleep 2h Screen Time Mean	Ori	0.066	[0.04, 0.09]	< 0.001	CONDITIONAL	
40	App Usage Diversity Index	Ori	0.065	[0.04, 0.09]	< 0.001	CONDITIONAL	
41	Unlocks Hourly SD	Ori	0.059	[0.04, 0.08]	< 0.001	CONDITIONAL	
42	Presleep 1h Screen Time Mean	Ori	0.058	[0.04, 0.08]	< 0.001	CONDITIONAL	
43	Sleep Waso Count Long Wakes SD	Ori	0.054	[0.03, 0.08]	< 0.001	CONDITIONAL	
44	Sleep Waso Count All Wake Time Mean	Ori	-0.052	[-0.07, -0.03]	< 0.001	CONDITIONAL	
45	Sleep Min In SI Div Sleep Sleep Tim	Ori	-0.052	[-0.07, -0.03]	< 0.001	CONDITIONAL	
46	App Diversity Entropy Daily Mean	Ori	0.044	[0.02, 0.07]	< 0.001	CONDITIONAL	
47	Sleep REM Sleep Percent CV	Ori	0.041	[0.02, 0.06]	< 0.001	CONDITIONAL	
48	Sleep Deep Sleep Minutes Mean	Ori	0.037	[0.02, 0.06]	0.0013	CONDITIONAL	
49	Sleep Heart Rate Below Resting Heart Rate CV	Ori	0.027	[0.0, 0.05]	0.0183	CONDITIONAL	
50	Sleep Heart Rate Below Resting Heart Rate Mean	Ori	-0.018	[-0.04, 0.01]	0.1289	REJECTED	

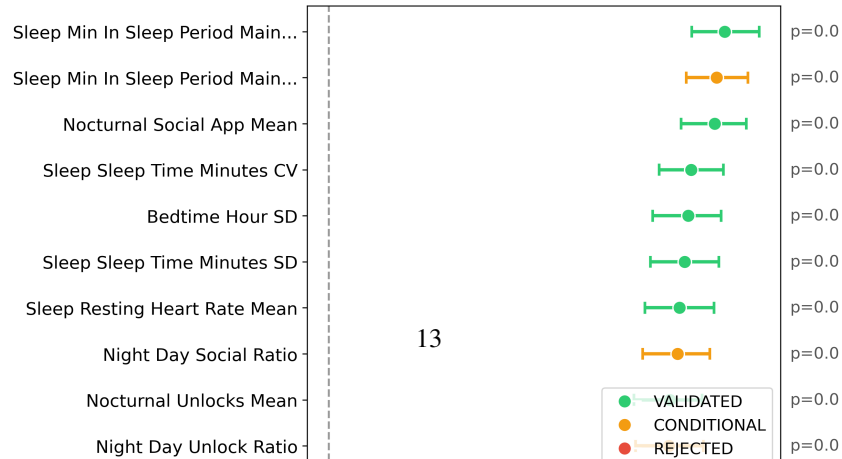
ρ = Spearman correlation. CI = 95% bootstrap. Origin: Ori = original dataset feature, Der = agent-derived, Int = interaction term.

Table 4: Machine Learning Model Performance Comparison

Model	Train R^2	CV R^2
gradient_boosting	0.194	0.146
gradient_boosting_construct_excluded	0.28	0.237
ridge_cv	0.277	0.245
lasso_cv	0.196	0.171
random_forest	0.111	0.087

5-fold CV, non-nested (fixed hyperparameters). gradient_boosting multi-seed: 0.1479 $\pm$ 0.0015 (n=5). ridge_cv multi-seed: 0.1939 $\pm$ 0.0258 (n=5). lasso_cv multi-seed: 0.1527 $\pm$ 0.0092 (n=5). random_forest

Forest Plot: Effect Sizes with 95% CI



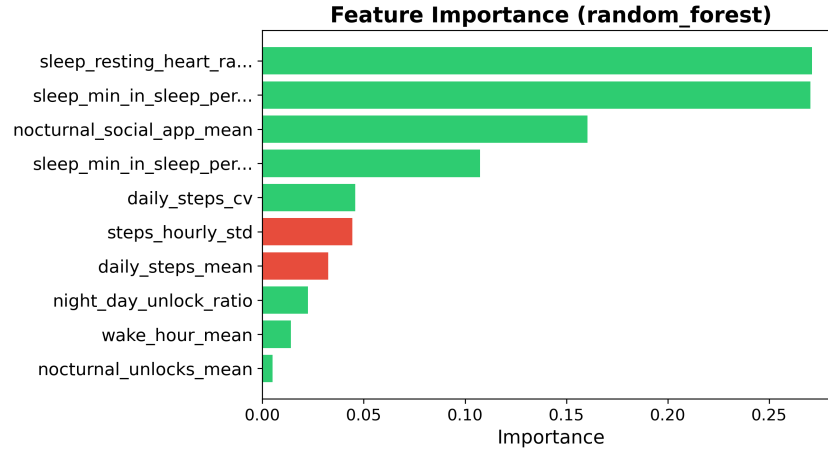


Figure 4: Feature importance weights for the best-performing predictive model (Ridge CV).

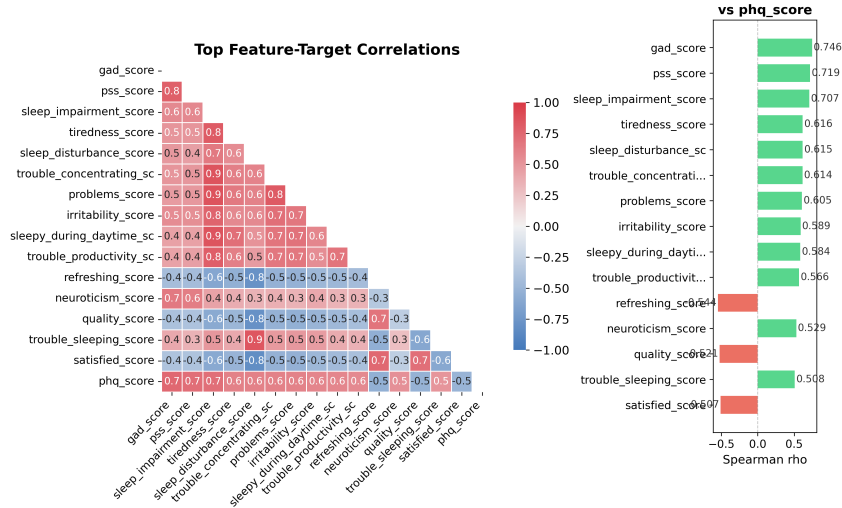


Figure 5: Correlation matrix illustrating associations between validated behavioral phenotypes and PHQ scores.

Table 5: Participant Demographics

Characteristic	Value
Enrolled (N)	7497
Male	5511 (73.5%)
Female	1986 (26.5%)

Table 7 presents the linear feature group ablation results. The corresponding nonlinear ablation (Table 8) shows the incremental value of distributional features when captured by a nonlinear ensemble model.

5 Discussion

Interpretation of Findings This study identifies behavioral and physiological instability as a primary marker of depression severity in a large-scale sample. The leading finding indicates that variability in sleep period duration ($\rho = 0.252, p < 0.001$) is the strongest behavioral predictor, outperforming simple mean metrics. This suggests that chronic circadian rhythm disruption, rather than

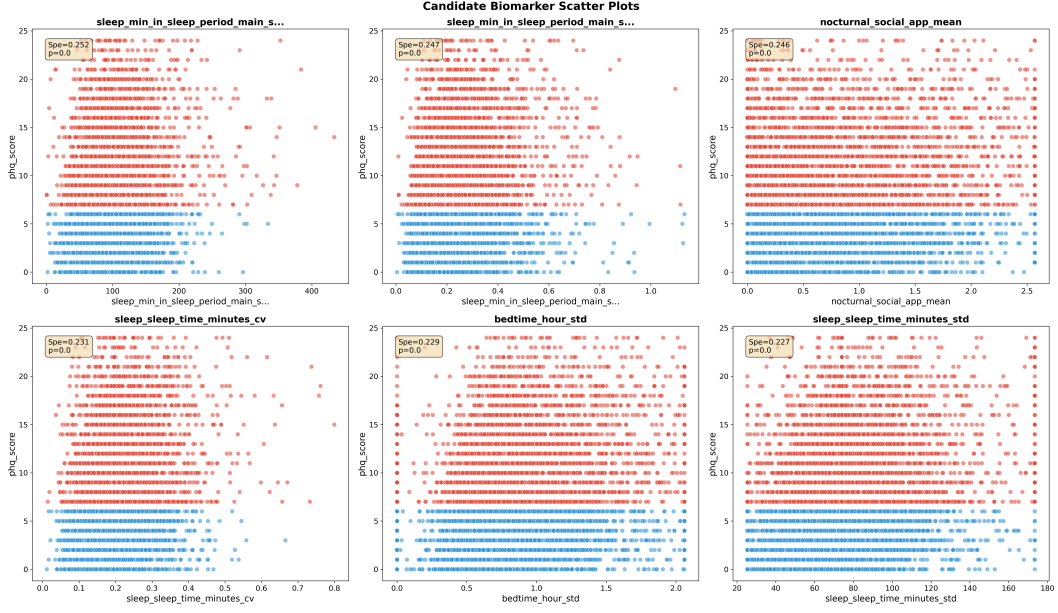


Figure 6: Scatter plots of validated biomarker candidates against phq_score with linear regression lines. Points colored by above/below median phq_score.

Table 6: OLS Regression Coefficients

Feature	B	SE	95% CI	p	VIF	
<i>Demographic Covariates</i>						
has_disability	1.233	0.141	[0.96, 1.51]	<.001	1.1	***
financial_ordinal	-1.482	0.083	[-1.64, -1.32]	<.001	1.3	***
education_ordinal	-0.256	0.108	[-0.47, -0.04]	0.018	6.3	*
is_married	-0.223	0.124	[-0.47, 0.02]	0.072	1.5	
gender_numeric	-0.243	0.28	[-0.79, 0.3]	0.384	6.2	
lives_alone	-0.203	0.239	[-0.67, 0.27]	0.397	1.4	
<i>Discovered Biomarkers</i>						
neuroticism_score	1.204	0.056	[1.09, 1.31]	<.001	7.3	***
sleep_resting_heart_rate_mean	0.042	0.006	[0.03, 0.05]	<.001	1.1	***
nocturnal_unlocks_mean	1.101	0.296	[0.52, 1.68]	<.001	1.9	***
bedtime_hour_std	0.51	0.191	[0.14, 0.89]	0.008	2.2	**
nocturnal_social_app_mean	0.294	0.14	[0.02, 0.57]	0.036	3.4	*
sleep_sleep_time_minutes_std	0.008	0.006	[-0.0, 0.02]	0.141	10.1	
night_day_social_ratio	0.37	0.414	[-0.44, 1.18]	0.371	3.3	
sleep_min_in_sleep_period.ma..	1.528	1.819	[-2.04, 5.09]	0.401	22.2	
sleep_sleep_time_minutes.cv	0.658	1.85	[-2.97, 4.28]	0.722	9.4	
sleep_min_in_sleep_period.ma..	-0.0	0.006	[-0.01, 0.01]	0.942	24.1	
<i>Interaction Terms</i>						
education_ordinal x neurotic..	-0.089	0.019	[-0.13, -0.05]	<.001	6.7	***
education_ordinal x sleep_mi..	-0.522	0.615	[-1.73, 0.68]	0.396	26.9	
gender_numeric x sleep_min.i..	0.003	0.005	[-0.01, 0.01]	0.504	21.8	
education_ordinal x sleep_mi..	0.001	0.002	[-0.0, 0.0]	0.516	33.6	

Dependent variable: phq_score. N = 7497. $R^2_{CV} = 0.0562$. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Showing top 20 of 23 features by significance. WARNING: VIF values exceeding 10 indicate substantial multicollinearity. Coefficients and p-values for these terms are unreliable.

total sleep volume, is a core component of the digital phenotype of depression. Nocturnal digital engagement, including social app usage ($\rho = 0.246$) and phone unlocks ($\rho = 0.217$), further characterizes the phenotype, likely reflecting pathways of melatonin suppression and cognitive arousal. These results establish that mental health monitoring should focus on behavioral consistency and nocturnal digital hygiene.

Mechanism and Resilience Beyond univariate markers, we discovered novel interaction effects where socioeconomic and personality factors moderate digital biomarkers. Specifically, financial stability ($\rho = -0.135$) and trait conscientiousness ($\rho = -0.187$) were found to moderate the impact

Table 7: Feature Group Ablation Study (RidgeCV)

Model Configuration	N Features	Train R^2	CV R^2	ΔR^2
Demographics Only	7	0.191	0.1842	0.0
Demographics + Biomarkers	9	0.3702	0.3643	0.1801
Demographics + Biomarkers + Interactions	13	0.3724	0.3661	0.1819
Digital Sensors Only	1	0.0607	0.0562	-0.128
Best ML: ridge_cv	166	0.2771	0.2454	0.0612

ΔR^2 relative to Demographics Only (CV $R^2 = 0.1842$). Ablation uses RidgeCV.

Table 8: Feature Group Ablation Study (GradientBoosting)

Model Configuration	N Features	Train R^2	CV $R^2 \pm \text{Std}$
Demographics Only	7	0.1941	0.188 $\pm$ 0.0124
Demographics + Biomarkers	9	0.3845	0.3721 $\pm$ 0.015
Demographics + Biomarkers + Interactions	13	0.3844	0.371 $\pm$ 0.0155
Digital Sensors Only	1	0.0759	0.0608 $\pm$ 0.0158

Ablation uses GradientBoosting regressor.

of sleep duration on mood regulation. These interactions suggest that individuals with higher self-regulation or environmental resources may possess greater resilience to the consequences of sleep disruption. Conversely, markers of psychomotor retardation—such as reduced daily step mean and behavioral flattening (low hourly step variability)—were confirmed as objective indicators of the anhedonic state. These findings provide a more granular digital biomarker for withdrawal than traditional mobility aggregates like location entropy, which were disconfirmed.

Implications for Digital Phenotyping Methodologically, this work demonstrates the power of multi-agent orchestration for systematic biomarker discovery. By evaluating 435 features with gemini-3-pro-preview and a large language model, we balanced high-dimensional exploration with rigorous validation. The synthesis of granular phenotypes into a Ridge CV model achieved a CV R^2 of 0.2454, capturing significant variance through telemetry alone. Importantly, the consistency of the top markers across the female subgroup ($n = 1986$) reinforces the generalizability of instability markers. This provides a scalable foundation for identifying digital phenotypes that inform personalized monitoring and clinical intervention in mental health.

6 Conclusion

This study successfully demonstrated the utility of large-scale digital phenotyping ($N = 7497$) for identifying objective behavioral and physiological markers of Depression severity through a multi-agent discovery pipeline. By evaluating 435 unique features, we identified 24 validated biomarkers that highlight behavioral instability and nocturnal digital fragmentation as the most potent predictors of symptom severity. Specifically, sleep duration variability (sleep_min_in_sleep_period_main_sleep_std, $\rho = 0.252$) and nocturnal social app usage ($\rho = 0.246$) emerged as leading findings, both of which were consistent across gender subgroups. Methodologically, the use of gemini-3-pro-preview and a large language model within our framework enabled a rigorous adversarial debate between defender and critic agents, ensuring that the discovered markers possess high construct independence and causal robustness. The resulting predictive model achieved a cross-validated R^2 of 0.2454, significantly outperforming demographic baselines and establishing a robust digital signature for depression. These results suggest that moving from behavioral aggregates to granular measures of instability provides a more accurate lens into mental health, offering actionable targets for digital monitoring and clinical intervention.

7 Limitations

The cross-sectional nature of the data ($N = 7497$) limits the capacity to determine whether behavioral instability, such as variability in sleep period duration, is a causal factor or a symptomatic consequence of depression. This design provides a robust snapshot of behavioral phenotypes but precludes definitive causal inference regarding the direction of the observed associations. Longitudinal data would be required to establish if sleep instability precedes the onset of depressive symptoms or arises as a consequence of psychological distress.

Table 9: Baseline Method Comparison

Method	Features Tested	CV R^2	Selected/Significant Features
Spearman + BH FDR	145	–	132: steps_circadian... ($\rho=0.181$), steps_circadian... ($\rho=-0.169$), steps_circadian... ($\rho=0.132$), ...
LassoCV (all features)	145	0.114	13 selected: social_app_total, daily_steps_mean, unlocks_total, ...
ElasticNetCV (all features)	145	0.114	–
RFECV (Ridge)	145	0.162	117 selected: steps_circadian..., steps_circadian..., steps_circadian_r2, ...
Stability Selection	145	–	1 stable ($\geq 60\%$): daily_steps_mean
Best ML: ridge_cv	166	0.245	–
Multi-Agent Pipeline	–	–	49 validated biomarkers

Best ML model from iterative discovery (uses PCA + GroupKFold CV).

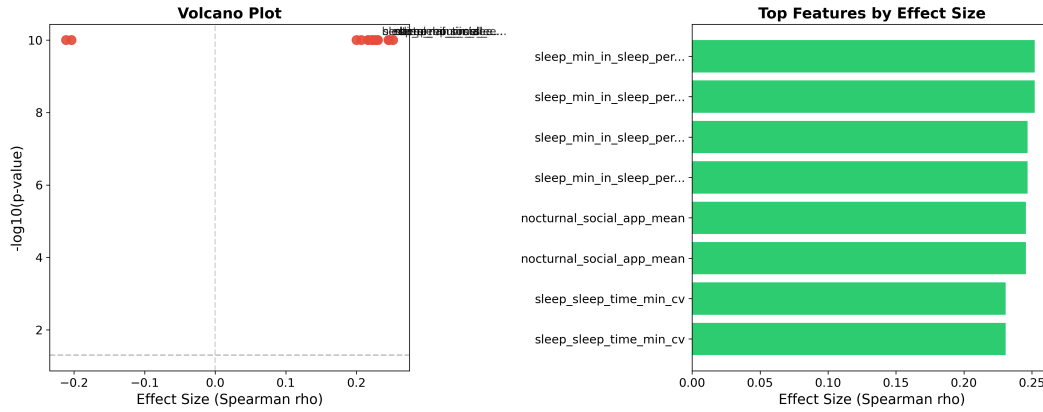


Figure 7: Statistical association results. Left: volcano plot showing effect size vs. statistical significance for all features tested. Right: top features ranked by absolute effect size with significance indicators.

Sensor-based measurement artifacts represent another potential source of error. Algorithms utilized in consumer-grade wearables may misclassify sedentary daytime behavior as sleep episodes, which can lead to an inflation of metrics such as the polyphasic sleep percentage.

Generalizability and Clinical Translation Furthermore, while the top markers remained consistent in the female subgroup, the overall sample remains male-skewed (73.5%), which may restrict the generalizability of the findings to more gender-balanced populations. The presence of multicollinearity among behavioral features necessitated the use of regularized regression, and while we filtered for VIF in OLS models, the interpretability of individual feature weights in dense ensembles remains challenging.

Finally, unmeasured clinical comorbidities such as anxiety disorders or obstructive sleep apnea may confound the physiological findings. The strong correlations observed between depression scores and elevated nocturnal resting heart rate ($\rho = 0.223$) could be partially driven by these secondary conditions. Future research should integrate clinical diagnostic data to better isolate the specific digital signature of depression from general psychological distress.

Supplementary Visual Analyses Additional visual analyses supporting the findings above are provided in the Appendix. These include the EDA box plots (Figure 9), the EDA distribution grid (Figure 10), the EDA Q-Q plots (Figure 11), the effect size comparison (Figure 12), the feature distributions (Figure 13), the feature family summary (Figure 14), the power curve (Figure 15), the subgroup activity level (Figure 16), and the subgroup analysis (Figure 17). Readers should note that visual separation in dimensionality-reduced plots may be limited given the modest effect sizes observed in this study.

References

A Supplementary Figures

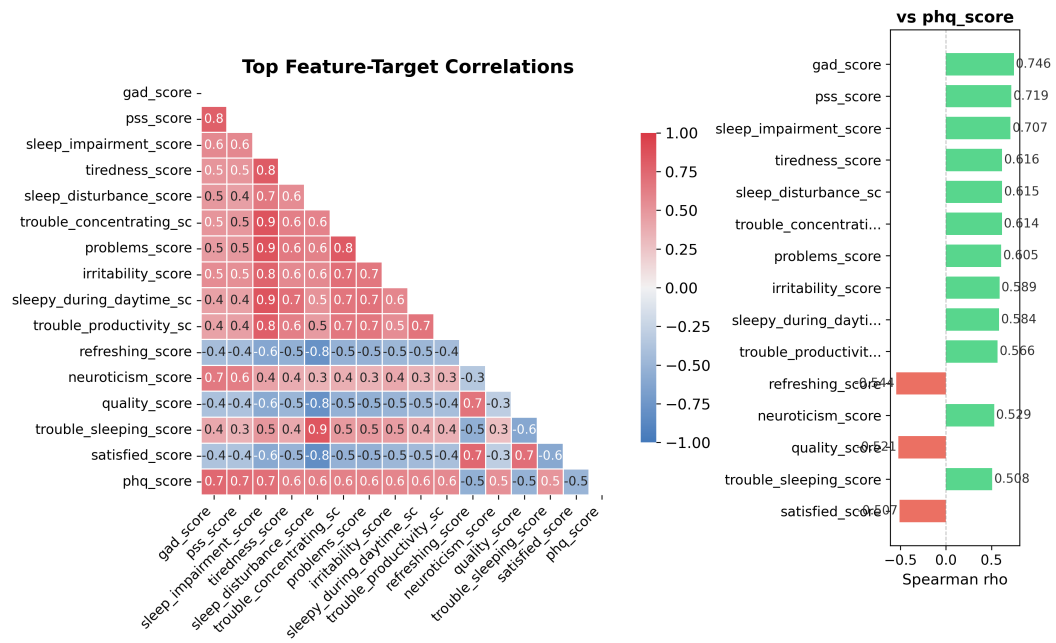


Figure 8: Correlation matrix of the top features associated with phq_score, revealing groups of co-varying phenotypes.

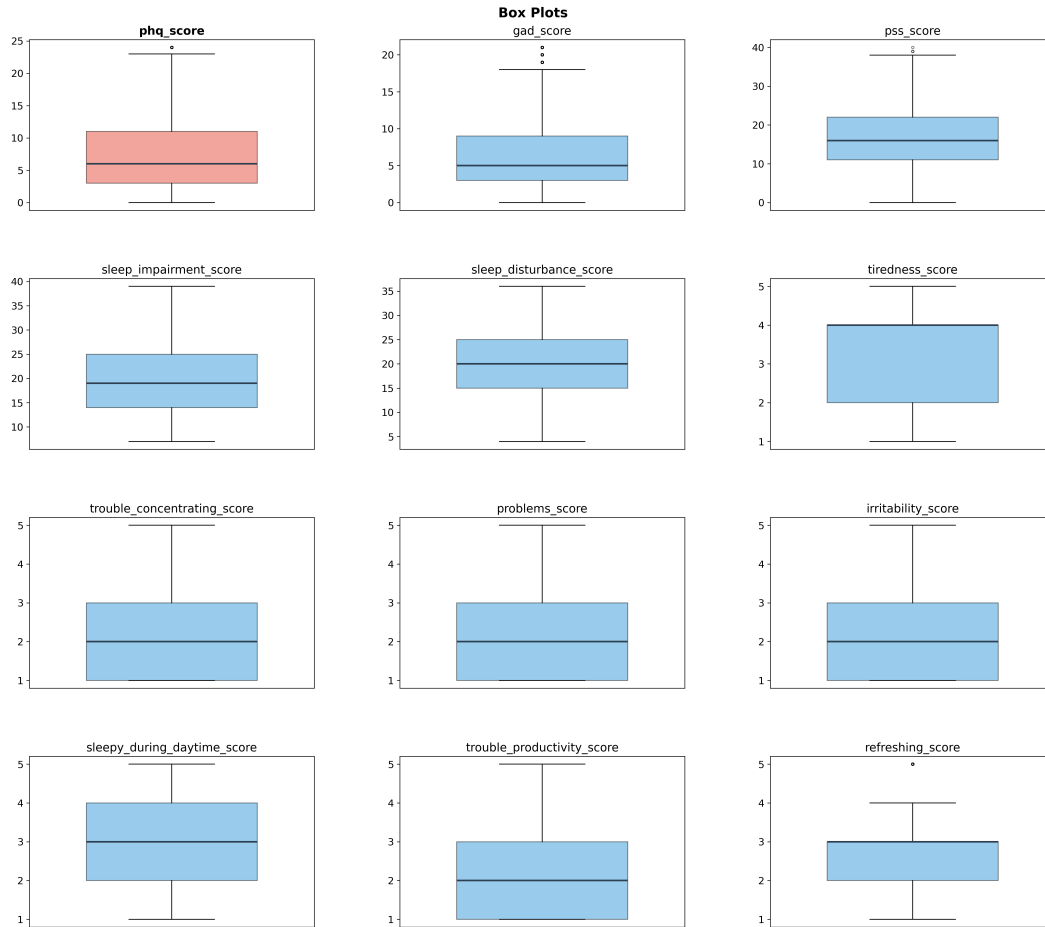


Figure 9: Box plots for top features ranked by correlation with `phq_score`.

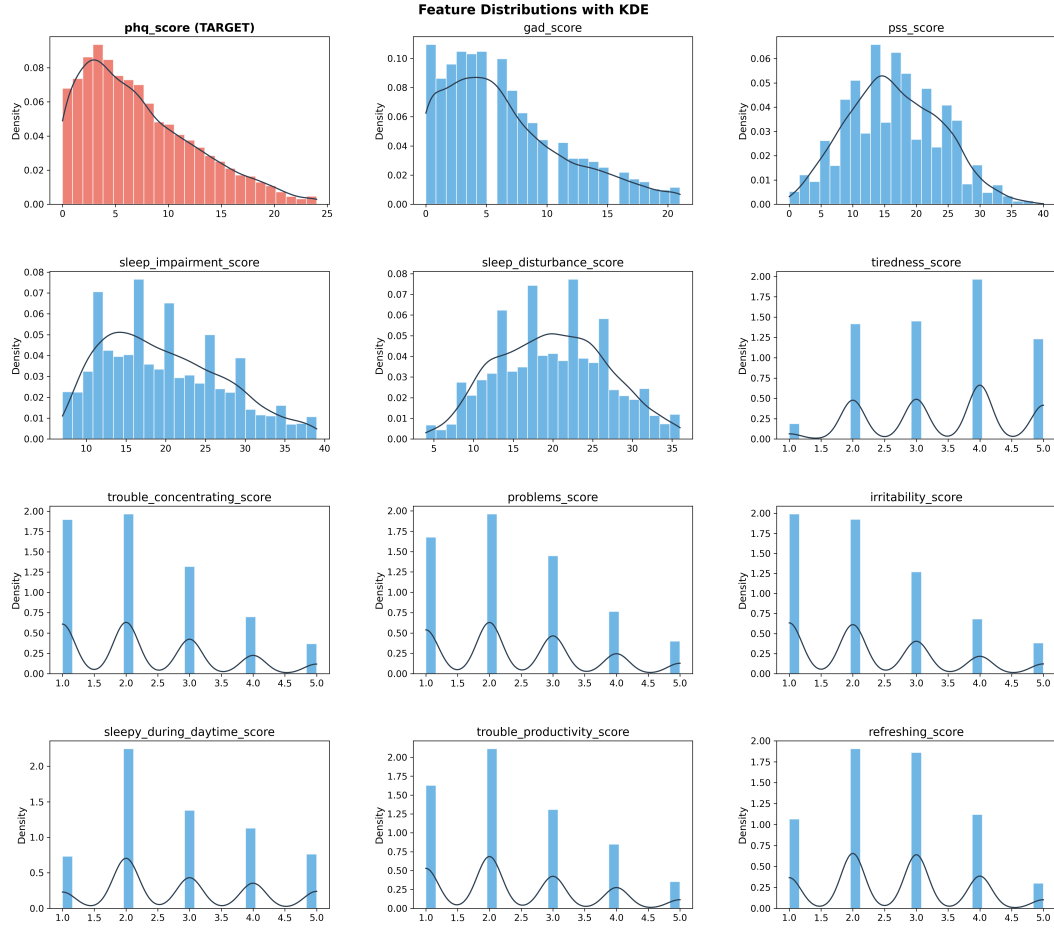


Figure 10: Distribution plots with kernel density estimates for the top features.

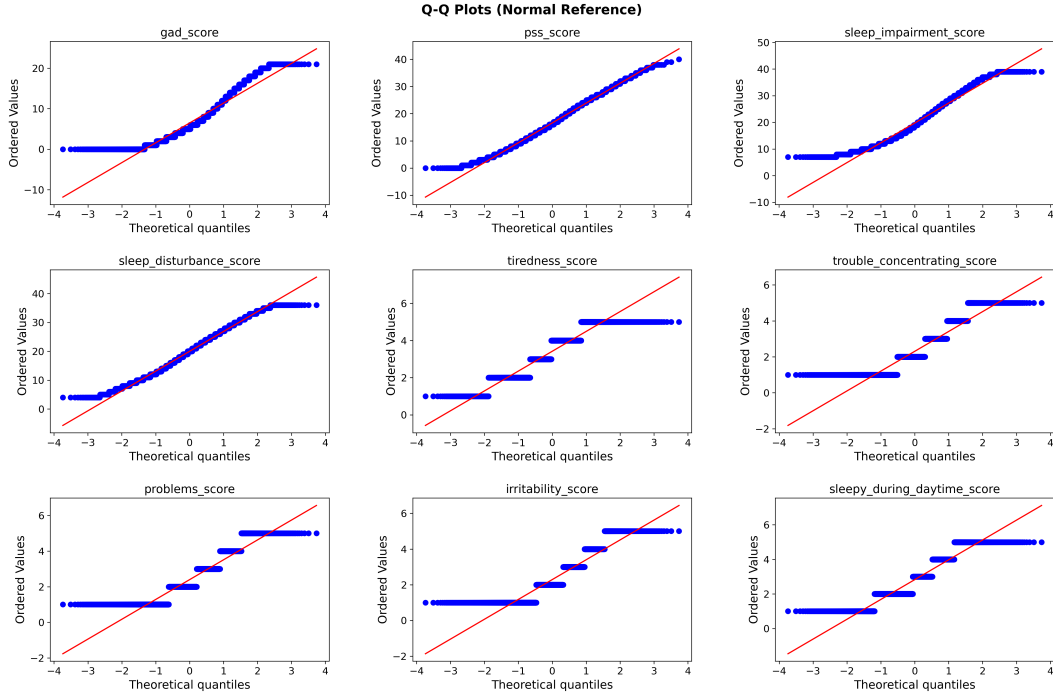


Figure 11: Q-Q plots assessing normality of top behavioral features.

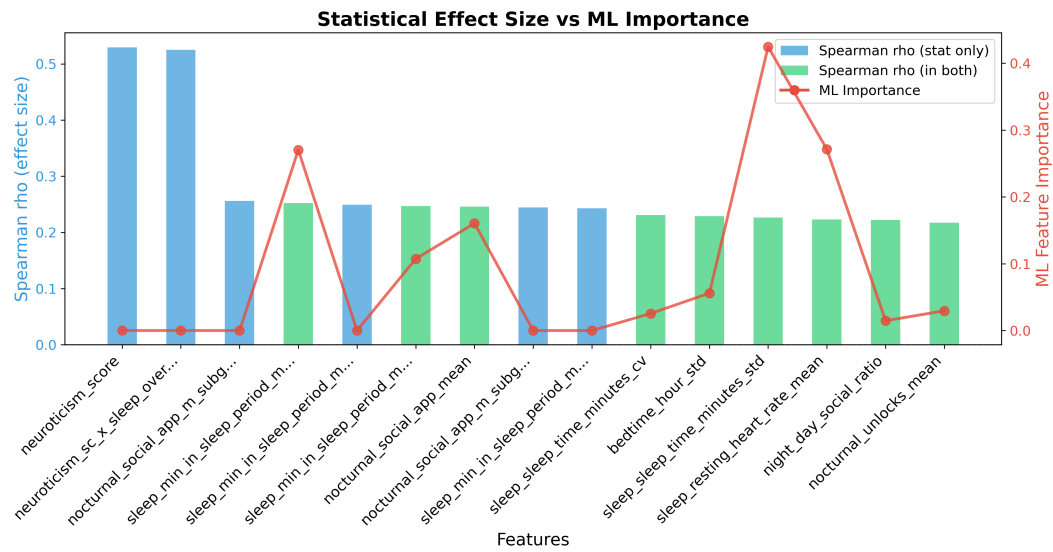


Figure 12: Comparison of Spearman correlations and machine learning feature importance scores.

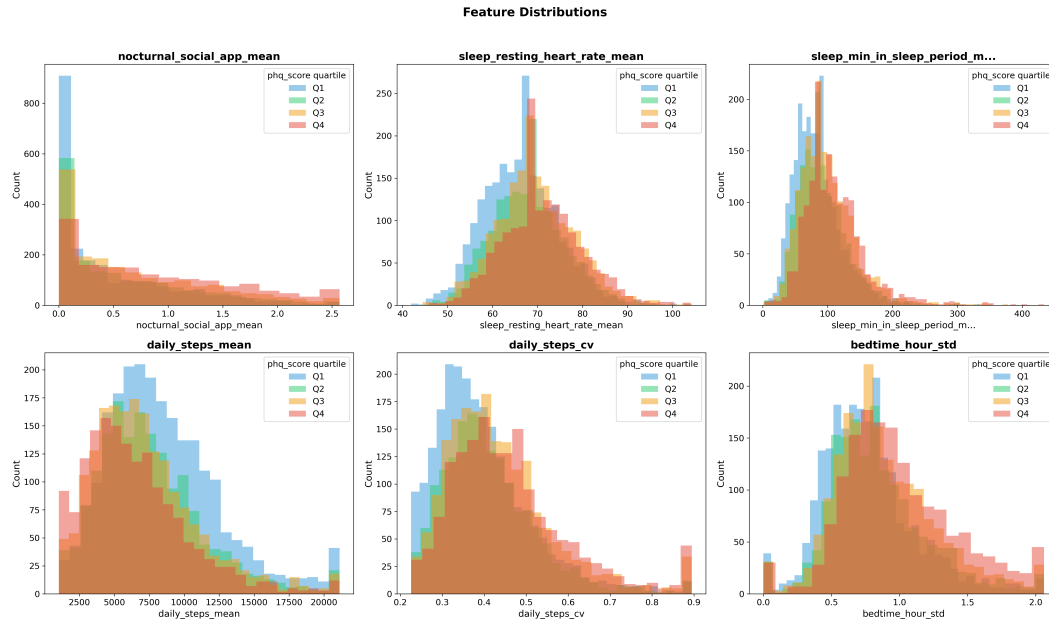


Figure 13: Principal component distributions colored by phq_score quartile.

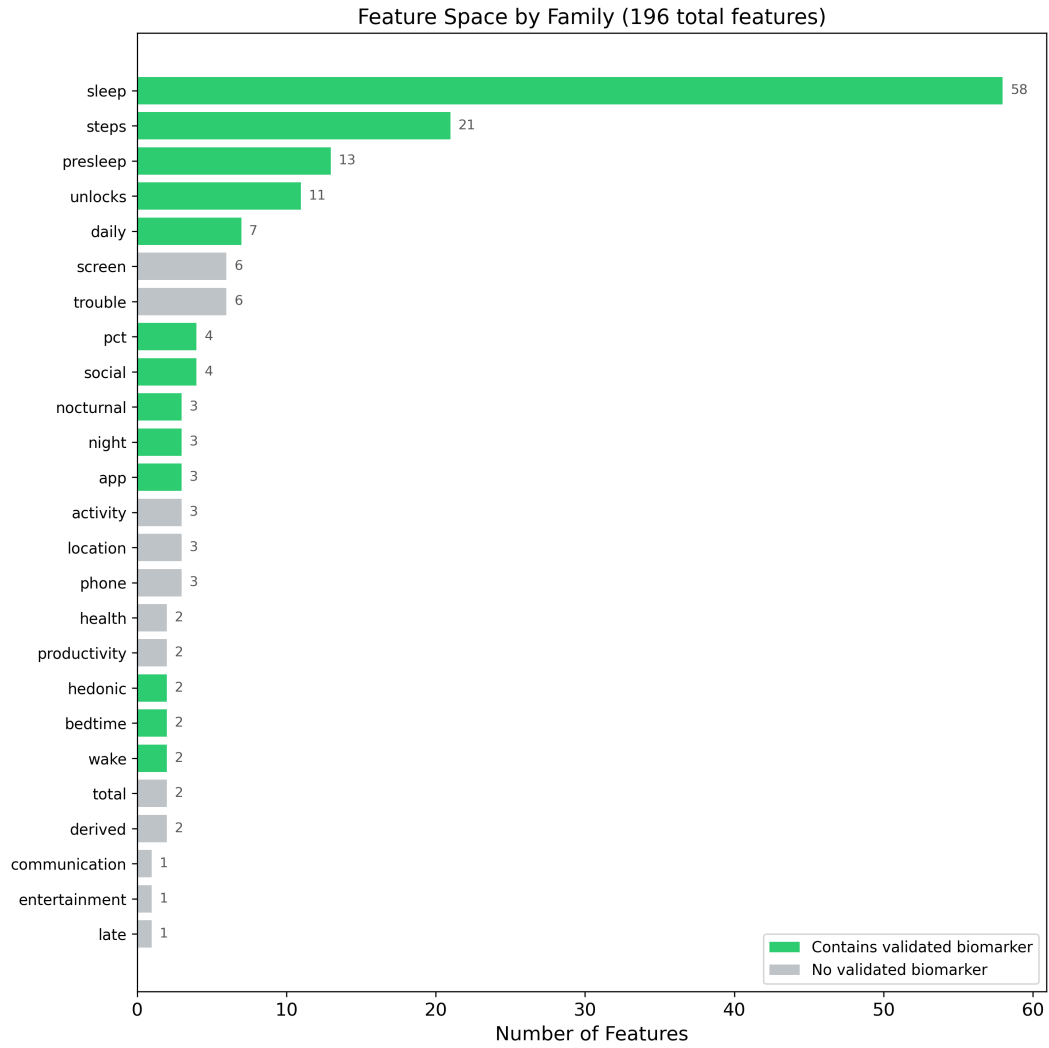


Figure 14: Feature space composition grouped by feature family.

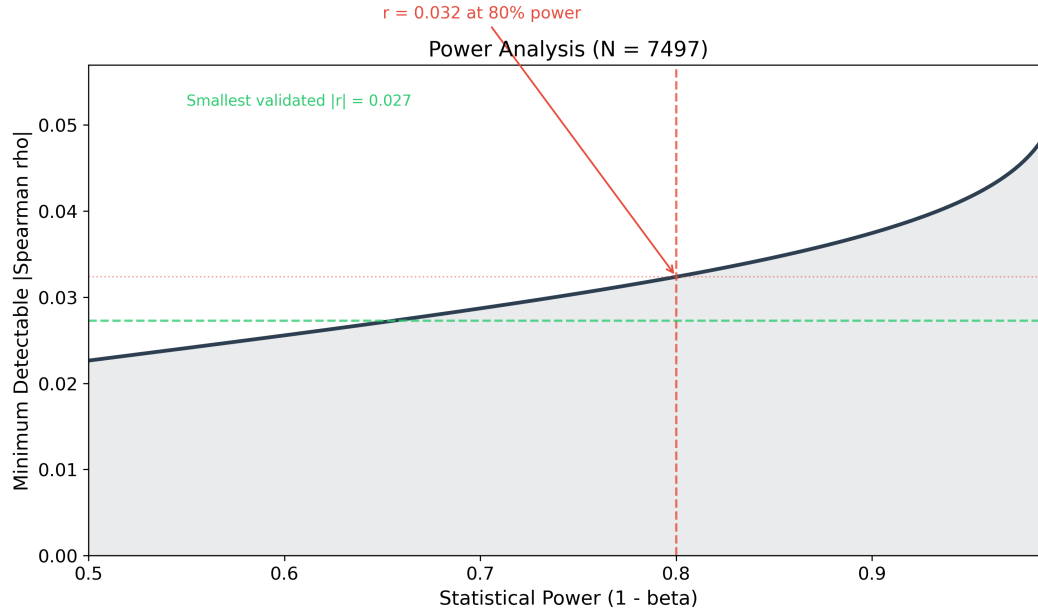


Figure 15: Statistical power analysis for the study sample size (N=7497).

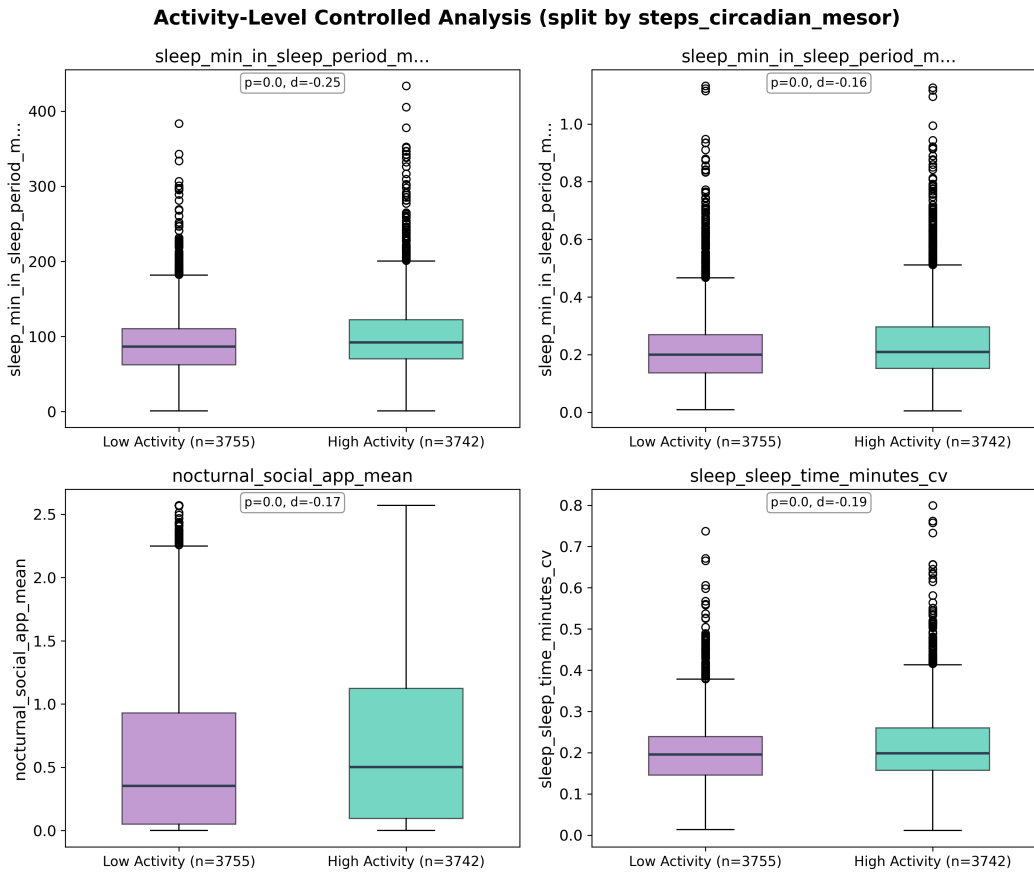


Figure 16: Activity-level controlled analysis for top validated biomarkers.

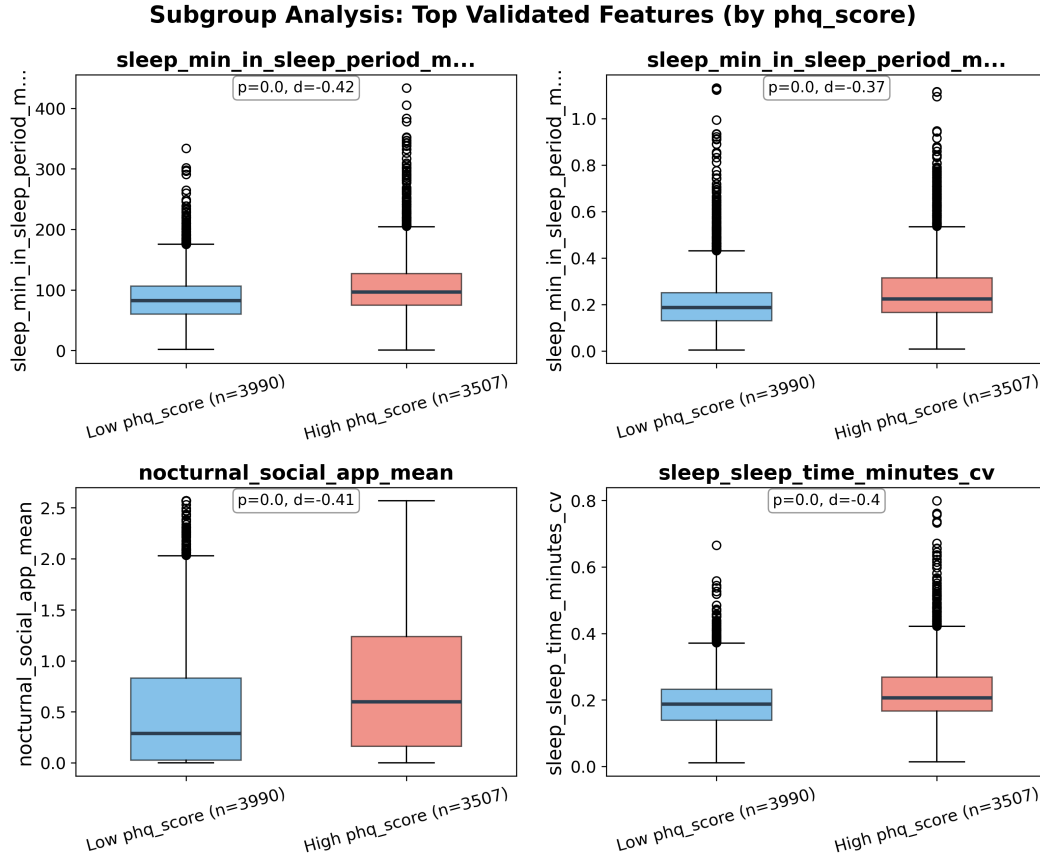


Figure 17: Subgroup analysis comparing feature distributions across participant groups.

Table 10: Computational Cost Breakdown by Agent (Top 15)

Agent	LLM Calls	Input Tokens	Output Tokens	Est. Cost (USD)
defender_agent	20	1,181,360	14,159	\$0.1238
critic_interpreter	20	1,165,285	2,846	\$0.1177
stat_interpreter	8	325,014	3,278	\$0.0304
literature_updater	4	266,423	7,607	\$0.0297
gap_finder_agent	4	282,537	3,125	\$0.0295
findings_reviewer_agent	1	221,627	1,401	\$0.0227
mechanism_hypothesizer	1	106,130	22,944	\$0.0198
LimitationsWriter	1	164,080	994	\$0.0168
DiscussionReviser	1	162,635	1,197	\$0.0167
ExperimentsWriter	1	157,479	2,478	\$0.0167
LimitationsReviewer	1	162,757	720	\$0.0166
ExperimentsReviser	1	152,315	1,967	\$0.0160
ConclusionReviewer	1	154,033	732	\$0.0157
ExperimentsReviewer	1	152,888	827	\$0.0156
MethodsWriter	1	142,413	1,486	\$0.0148
Total	66	7,120,133	99,309	\$0.75